IMAT 모의고사 9회

60 questions · 100 minutes · +1.5 / −0.4 / 0

Section 1 — Reading Skills & General Knowledge (Q1–4)

Passage for Q1–2

> Many countries have introduced taxes on sugar-sweetened beverages. Early evaluations show that purchases of taxed drinks fell, and some manufacturers reduced the sugar content of their products to avoid the tax. Whether these changes translate into lower rates of obesity is harder to establish, because obesity develops slowly and is influenced by many factors. Critics also argue that such taxes fall more heavily on low-income households, although supporters reply that those same households bear a disproportionate share of diet-related disease.

1. According to the passage, which effect of sugar taxes is most clearly established?

A. Elimination of diet-related disease
B. Higher incomes for poorer households
C. A large fall in obesity rates
D. An increase in sugar content
E. A reduction in purchases of taxed drinks

2. How do supporters respond to the criticism that the taxes affect low-income households more?

A. They argue that poorer households also suffer more from diet-related illness and thus may benefit more.
B. They deny that poorer households buy sugary drinks.
C. They claim the tax has no effect on prices.
D. They propose abolishing the tax.
E. They say obesity is not influenced by diet.

3. Who proposed the theory of evolution by natural selection, published in *On the Origin of Species* (1859)?

A. Gregor Mendel
B. Jean-Baptiste Lamarck
C. Carl Linnaeus
D. Charles Darwin
E. Thomas Malthus

4. Which body of the European Union is directly elected by citizens of member states?

A. The Court of Justice
B. The European Parliament
C. The European Commission
D. The European Council
E. The European Central Bank

Section 2 — Logical Reasoning & Problem Solving (Q5–9)

5. "Either the sample was contaminated, or the machine is faulty. The machine is not faulty." Therefore:

A. Both are true.
B. Nothing follows.
C. Neither is true.
D. The sample was contaminated.
E. The sample was not contaminated.

6. A tablet contains 250 mg of active ingredient. A patient needs 1.5 g per day, divided equally into 3 doses. How many tablets per dose?

A. 2
B. 6
C. 3
D. 4
E. 1

7. In a tournament, each of 6 teams plays every other team exactly once. How many matches are played?

A. 15
B. 12
C. 36
D. 6
E. 30

8. If all Blips are Zogs and some Zogs are Fems, which statement must be true?

A. All Fems are Zogs.
B. No Blips are Fems.
C. All Blips are Fems.
D. Some Zogs are Blips (assuming Blips exist).
E. Some Blips are Fems.

9. The ratio of boys to girls in a class is 3 : 4. If there are 12 boys, how many students are there in total?

A. 16
B. 24
C. 21
D. 36
E. 28

Section 3 — Biology (Q10–32)

10. Which organelle is responsible for the detoxification of drugs in liver cells?

A. Smooth ER
B. Rough ER
C. Centrosome
D. Ribosome
E. Nucleolus

11. Which enzyme unwinds the DNA double helix at the replication fork?
A. Primase
B. DNA polymerase
C. Telomerase
D. Ligase
E. Helicase

12. During which phase does the nuclear envelope reform around the chromosomes?
A. S phase
B. Metaphase
C. Telophase
D. Prophase
E. Anaphase

13. Which of the following yields energy in the form of ATP during glycolysis?
A. The Calvin cycle
B. Photophosphorylation
C. β-oxidation
D. Substrate-level phosphorylation
E. Oxidative phosphorylation

14. Achondroplasia is autosomal dominant. An affected heterozygous man and an unaffected woman have a child. What is the probability that the child is affected?
A. 3/4
B. 1/2
C. 1
D. 0
E. 1/4

15. Which of the following is the correct sequence in a food chain?
A. Decomposer → producer → herbivore
B. Secondary consumer → producer → primary consumer
C. Producer → primary consumer → secondary consumer
D. Herbivore → producer → carnivore
E. Carnivore → herbivore → producer

16. Which muscle is the main muscle of inspiration?
A. Diaphragm
B. Internal intercostals
C. Abdominal muscles
D. Biceps
E. Trapezius

17. Which of the following carbohydrates forms the structural component of plant cell walls?
A. Cellulose
B. Lactose
C. Starch
D. Chitin
E. Glycogen

18. Which statement about the autonomic nervous system is correct?

A. The parasympathetic system prepares the body for "fight or flight".
B. The sympathetic system decreases heart rate.
C. The sympathetic system dilates the pupils and increases heart rate.
D. The autonomic system controls skeletal muscles voluntarily.
E. The parasympathetic system increases blood glucose.

19. Systemic blood pressure is highest in the:

A. capillaries.
B. venules.
C. veins.
D. vena cava.
E. aorta.

20. A cross produces offspring in a 9 : 3 : 4 ratio. This is an example of:

A. sex linkage.
B. codominance.
C. incomplete dominance.
D. epistasis.
E. linkage without crossing over.

21. Which hormone is produced by the corpus luteum after ovulation?

A. FSH
B. Progesterone
C. LH
D. Prolactin
E. Oxytocin

22. Which of the following is a purine?

A. Cytosine
B. Ribose
C. Uracil
D. Thymine
E. Adenine

23. Which antibody class is the first to be produced in a primary immune response?

A. IgE
B. IgD
C. IgG
D. IgA
E. IgM

24. Which of the following substances is secreted (rather than filtered) into the tubule in the kidney?

A. Glucose
B. Red blood cells
C. Albumin
D. H^+ ions
E. Amino acids

25. How many genetically different gametes can an individual with genotype AaBbCcDd produce (independent assortment)?

A. 16
B. 8
C. 32
D. 4
E. 64

26. An enzyme's Vmax is reached when:

A. a competitive inhibitor is added.
B. the temperature is 0 °C.
C. the enzyme is denatured.
D. the substrate concentration is zero.
E. all active sites are saturated with substrate.

27. Which type of mutation involves a segment of chromosome breaking off and reattaching in reversed orientation?

A. Duplication
B. Deletion
C. Inversion
D. Substitution
E. Translocation

28. Plant cells placed in a hypotonic solution become:

A. crenated.
B. plasmolysed.
C. turgid.
D. lysed.
E. flaccid.

29. Which of the following is a eukaryotic pathogen?

A. Streptococcus pyogenes
B. Plasmodium falciparum
C. Influenza virus
D. HIV
E. Escherichia coli

30. The villi of the small intestine increase:

A. peristalsis in the oesophagus.
B. acid production.
C. the surface area for absorption.
D. the time food spends in the stomach.
E. bile production.

31. How many different codons can be formed from the four RNA bases?

A. 4
B. 64
C. 20
D. 16
E. 61

32. Which of the following is an example of convergent evolution?
A. Darwin's finches
B. Homologous genes in mammals
C. Antibiotic resistance
D. The wings of bats and the wings of insects
E. The forelimbs of humans and whales

Section 4 — Chemistry (Q33–47)

33. Which sample contains the greatest number of atoms?
A. 1 mol of H_2O
B. 1 mol of CH_4
C. 1 mol of NaCl
D. 1 mol of He
E. 1 mol of O_2

34. In the reaction $Cl_2 + 2NaBr \rightarrow 2NaCl + Br_2$, which statement is correct?
A. Cl_2 is oxidised.
B. No redox occurs.
C. Br^- is reduced.
D. Cl_2 is the oxidising agent.
E. Na^+ is oxidised.

35. What is the pH of a 0.005 mol/L solution of H_2SO_4 (assume complete dissociation of both protons)?
A. 3
B. 2.3
C. 2
D. 1
E. 5

36. Which of the following groups contains the alkaline earth metals?
A. Group 17
B. Group 2
C. Group 13
D. Group 18
E. Group 1

37. How many lone pairs are on the central atom of H_2O?
A. 1
B. 0
C. 3
D. 4
E. 2

38. 5.6 g of iron (Fe = 56) reacts completely with sulfur: Fe + S → FeS. What mass of FeS (M = 88 g/mol) is formed?

A. 5.6 g
B. 8.8 g
C. 3.2 g
D. 88 g
E. 17.6 g

39. Which compound will react with sodium hydrogencarbonate to release CO_2?

A. Ethanal
B. Propanone
C. Ethanol
D. Ethane
E. Ethanoic acid

40. How many moles of NaOH are needed to neutralise 50 mL of 0.2 mol/L H_2SO_4?

A. 0.04
B. 0.02
C. 0.1
D. 0.005
E. 0.01

41. Which of the following has the lowest boiling point?

A. CH_4
B. C_2H_5OH
C. HF
D. H_2O
E. NH_3

42. Which of the following solutions is neutral at 25 °C?

A. $[H^+] = 10^{-5}$ mol/L
B. pOH = 4
C. pH = 9
D. $[OH^-] = 10^{-3}$ mol/L
E. $[H^+] = 10^{-7}$ mol/L

43. The half-life of a first-order reaction:

A. is independent of the initial concentration.
B. depends on the initial concentration.
C. is zero.
D. doubles each time.
E. is always 1 s.

44. What type of isomerism is shown by the two forms of lactic acid (2-hydroxypropanoic acid)?

A. No isomerism
B. Optical isomerism
C. Chain isomerism
D. Structural isomerism
E. Geometric (cis–trans) isomerism

45. Which element has the highest electronegativity?
A. C
B. O
C. F
D. Cl
E. N

46. How many electrons are transferred when one Fe^{3+} ion is reduced to Fe?
A. 6
B. 1
C. 3
D. 2
E. 4

47. Which reaction is characteristic of alkenes but not alkanes?
A. Cracking
B. Dissolving in water
C. Electrophilic addition of Br_2 (decolourising bromine water)
D. Combustion
E. Free-radical substitution

Section 5 — Physics & Mathematics (Q48–60)

48. The velocity–time graph of an object is a straight line from 0 to 12 m/s over 4 s. What is the distance travelled?
A. 48 m
B. 12 m
C. 24 m
D. 6 m
E. 3 m

49. A pump does 6000 J of useful work while consuming 8000 J of energy. What is its efficiency?
A. 133%
B. 80%
C. 25%
D. 75%
E. 60%

50. Two identical 10 Ω resistors are connected in parallel to a 5 V supply. What is the total current?
A. 0.5 A
B. 2 A
C. 5 A
D. 0.25 A
E. 1 A

51. A note has a frequency of 440 Hz. What is its period?

A. 0.00227 s
B. 440 s
C. 2.27 s
D. 0.227 s
E. 0.0227 s

52. A block of volume 0.002 m^3 is fully submerged in water. What is the buoyant force? (ρ = 1000 kg/m^3, g = 10 m/s^2)

A. 2 N
B. 2000 N
C. 20 N
D. 10 N
E. 200 N

53. A 70 kg astronaut on the Moon (g = 1.6 m/s^2) has a weight of:

A. 70 N
B. 700 N
C. 43.75 N
D. 112 N
E. 1120 N

54. The temperature 27 °C is equal to:

A. 327 K
B. 300 K
C. 273 K
D. 246 K
E. 27 K

55. If log_a(8) = 3, what is a?

A. 24
B. 1/2
C. 8/3
D. 3
E. 2

56. For which value of x in [0°, 360°) is sin x = −1?

A. 270°
B. 180°
C. 360°
D. 90°
E. 0°

57. Two cards are drawn without replacement from a standard 52-card deck. What is the probability that both are aces?

A. 1/221
B. 1/169
C. 1/26
D. 4/52
E. 1/13

58. What is the gradient of the line $4x + 2y = 6$?

A. 2
B. -2
C. 3
D. $-1/2$
E. 4

59. If $x = 2$ is a root of $x^2 + kx - 10 = 0$, what is k?

A. 7
B. 2
C. 5
D. 3
E. -3

60. Which of the following is a vector quantity?

A. Temperature
B. Speed
C. Energy
D. Displacement
E. Mass

정답 및 해설

번호	정답	해설
1	E	과세 음료 구매 감소가 확인된 효과.
2	A	저소득층이 식이 관련 질환 부담도 더 크므로 더 큰 이득.
3	D	다윈.
4	B	유럽의회.
5	D	선언 삼단논법(선언지 제거).
6	A	1500/3 = 500 mg → 2정.
7	A	${}_6C_2 = 15$.
8	D	Blip ⊂ Zog → Blip이 존재하면 일부 Zog는 Blip.
9	E	12 → 3배 → 3+4 = 7 × 4 = 28.
10	A	활면소포체(시토크롬 P450).
11	E	헬리케이스.
12	C	말기.
13	D	기질 수준 인산화.
14	B	Aa × aa → 1/2.
15	C	생산자 → 1차 소비자 → 2차 소비자.
16	A	횡격막.
17	A	셀룰로스.
18	C	교감: 동공 확대, 심박↑.
19	E	대동맥.
20	D	열성 상위(9:3:4).
21	B	프로게스테론.
22	E	아데닌(퓨린: A, G).
23	E	IgM.
24	D	H^+ 분비.
25	A	$2^4 = 16$.
26	E	활성부위 포화.
27	C	역위.
28	C	팽압 상태(turgid).
29	B	말라리아원충(원생생물).
30	C	흡수 표면적 증가.
31	B	$4^3 = 64$.
32	D	박쥐 날개 vs 곤충 날개(상사기관).
33	B	CH_4 원자 5개/분자.
34	D	Cl_2 환원 → 산화제.
35	C	$[H^+] = 0.01$ → pH 2.

번호	정답	해설
36	B	2족.
37	E	2개.
38	B	0.1 mol → 8.8 g.
39	E	카복실산 + $NaHCO_3$ → CO_2.
40	B	H_2SO_4 0.01 mol × 2 = 0.02 mol.
41	A	CH_4(수소결합 없음, 분자량 작음).
42	E	$[H^+] = 10^{-7}$.
43	A	$t_{½} = \ln 2/k$, 초기 농도 무관.
44	B	키랄 탄소 → 광학 이성질.
45	C	플루오린.
46	C	3개.
47	C	브로민수 탈색(친전자성 첨가).
48	C	삼각형 면적 ½×4×12 = 24 m.
49	D	6000/8000 = 75%.
50	E	5 Ω → 1 A.
51	A	1/440 ≈ 0.00227 s.
52	C	ρVg = 1000×0.002×10 = 20 N.
53	D	70 × 1.6 = 112 N.
54	B	27 + 273 = 300 K.
55	E	$2^3 = 8$.
56	A	270°.
57	A	4/52 × 3/51 = 12/2652 = 1/221.
58	B	$y = -2x + 3$.
59	D	$4 + 2k - 10 = 0 \rightarrow k = 3$.
60	D	변위.